

An Efficient Blood-Cell Segmentation for the Detection of Hematological Disorders

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Abstract—The automatic segmentation of blood cells for detecting hematological disorders is a crucial job. It has a vital role in diagnosis, treatment planning, and output evaluation. The existing methods suffer from the issues like noise, improper seed-point detection, and oversegmentation problems, which are solved here using a Laplacian-of-Gaussian (LoG)-based modified highboosting operation, bounded opening followed by fast radial symmetry (BOFRS)-based seed-point detection, and hybrid ellipse fitting (EF), respectively. This article proposes a novel hybrid EF-based blood-cell segmentation approach, which may be used for detecting various hematological disorders. Our prime contributions are: 1) more accurate seed-point detection based on BO-FRS; 2) a novel least-squares (LS)-based geometric EF approach; and 3) an improved segmentation performance by employing a hybridized version of geometric and algebraic EF techniques retaining the benefits of both approaches. It is a computationally efficient approach since it hybridizes noniterative-geometric and algebraic methods. Moreover, we propose to estimate the minor and major axes based on the residue and residue offset factors. The residue offset parameter, proposed here, yields more accurate segmentation with proper EF. Our method is compared with the state-of-the-art methods. It outperforms the existing EF techniques in terms of dice similarity, Jaccard score, precision, and F1 score. It may be useful for other medical and cybernetics applications.

Index Terms—Acute lymphoblastic leukemia (ALL), acute myeloid leukemia (AML), ellipse fitting (EF), hematological disorder, segmentation, sickle cell anemia (SCA).

I. INTRODUCTION

AUTOMATIC segmentation is a significant task in image processing applications [1], [2], [4]–[19], [39], [40]. It has an important role in the detection, classification, and diagnosis of hematological disorders: sickle cell anemia (SCA), acute lymphoblastic leukemia (ALL), and acute myeloid

leukemia (AML). SCA causes significant morphological variations of red blood cells (RBCs) [1], [9]. However, AML and ALL affect white blood cells (WBCs) [3], [34]. Hence, the analysis of cell morphology has a vital role in the diagnosis of hematological disorders [1], [20].

Manual counting and segmentation of cells are time consuming and truly are a more challenging job [1], [9], [18]. The presence of noise, unwanted cells, overlapped cells, and poor contrast make this task more difficult. On the other hand, automatic segmentation enhances the robustness and, thus, makes diagnosis and treatment planning more accurate [1], [9]. Proper segmentation of overlapped or touched cells enhances detection and classification performance. Many researchers have currently recommended deep-learning-based schemes to yield promising segmentation and classification in the medical imaging and computer vision field [37], [38], [40]–[42]. However, traditional deep learning methods are computationally expensive and require vast datasets for training [39], [43]–[45].

Recently, ellipse fitting (EF) has become a preferred approach for the segmentation of overlapping objects cells [1], [5]–[13]. González-Hidalgo *et al.* [9] concentrated on concave point identification, followed by an ellipse adjustment technique to segment overlapped RBCs effectively. However, this method [9] is unable to segment an ambiguously overlapping-object precisely since it is impossible to identify intersection-points. On the other hand, Li *et al.* [8] have suggested an efficient EF-based approach to estimate fetal head circumference in ultrasound images automatically. However, it may demonstrate poor performance due to false localization of region of interest (ROI), missing of ROI, or inefficient EF. False localization of ROI occurs due to structural similarity between the head circumference and the background surface. On the other hand, low contrast of the head circumference results in missing ROI, whereas missing the skull's boundaries yields inefficient EF.

Azhar and Tjahjadi [10] have presented an EF and contour-movement-based method to classify a human body-posture. Here, the EF approach is employed on the inverted-pendulum-human-body model to classify the body posture. However, this EF approach is unable to detect the contour points of the human body. It only gives information about the global angle. On the other hand, the contour moment is estimated to determine the human body's angle concerning the vertical axis. Chen and Epps [12] have suggested a convex-hull and EF-based approach to extract the pupil boundary of the eye more

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TABLE I
ACRONYM LIST

ALL	Acute Lymphoblastic Leukemia
AML	Acute Myeloid Leukemia
BO-FRS	Bounded Opening followed by Fast Radial Symmetry
BE-FRS	Bounded Erosion-Fast Radial Symmetry
DSC	Dice Similarity Coefficient
EFM	Ellipse Fitting Method
FRS	Fast Radial Symmetry
LoG	Laplacian-of-Gaussian
LS	Least Square
RBC	Red Blood Cell
SCA	Sickle Cell Anemia
WBC	White Blood Cell

accurately. They have employed a least-square (LS)-based EF approach to evaluate the pupil size efficiently. However, this EF technique depicts poor performance in some specific situations like when a major part of the pupil is occluded.

Prasad *et al.* [6] have presented an elegant LS-based EF technique. It is a noniterative geometric ellipse fitting approach. It is stable and computationally efficient also. It has comparatively better selectivity to an elliptic curve than many current techniques, since it is a geometric-based EF approach; not an algebraic-based approach. The results presented in [6] signify that it delivers better performance than [24]. However, it [6] demonstrates weak performance, particularly in the presence of inflection points in the arc [36]. It may also yield poor performance in the presence of non-Gaussian outliers since it is an LS-based approach.

In 2015, Zafari *et al.* [5] presented an efficient EF method (EFM) to segregate overlapping objects for silhouette images. They have successfully extracted seed points by applying bounded erosion (BE) [5], [21], and FRS [22]. Then, edge and seed point information has been integrated to find out contour evidence. Finally, an EF technique [23] is applied to estimate contour and segment overlapped objects as well [5]. They focus on the estimation of the proper eigenvector. Simplicity and computational efficiency are the salient features of the algebraic EF approach. The major limitations of this method are as follows.

- 1) It may display poor performance in the presence of non-Gaussian outliers due to low breakdown-points problem since it is an LS-based approach.
- 2) Moreover, all the eigenvalues become zero when the data points perfectly lie on the ellipse, for which it provides no solution.

In 2020, Meng *et al.* [7] presented a novel arc adjacency matrix-based EF technique. In this approach, diagram-based arc adjacency is estimated from the elliptic arcs. Then, curvature and region constraints are applied to make it sparse. Finally, effective EF is achieved depending on twice the eigen-decomposition of cumulative matrices based on the Jacobi approach. The limitations of this method are as follows.

- 1) It may consume a lot of time for the validation of candidate combinations [7]. The validation is employed to decide whether a candidate combination can yield a real ellipse or not.
- 2) It cannot detect small ellipses efficiently due to the unavailability of sufficient curvature information.

- 3) Its recall performance degrades due to the presence of occluded ellipses in an image.

Thus, it is observed that the existing methods suffer from the problems of noise, intensity inhomogeneity, unwanted similar objects, and oversegmentation. Therefore, the authors are motivated to propose a novel hybrid EF-based blood-cell segmentation technique to mitigate the above issues and yield more precise segmentation with proper EF, which may be employed for blood cell segmentation to detect hematological disorders. There is a strong need for a new theoretical investigation in this context. This article proposes a new theoretical investigation on the estimation of major and minor axes of an ellipse based on residue and residue offset. The acronym list is demonstrated in Table I.

This article proposes a novel hybrid EF-based blood-cell segmentation technique for the detection of hematological disorders. The proposed method emphasizes the elimination of unwanted cells (WBCs and platelets in SCA detection; RBCs and platelets in AML and ALL detection) using the k-means clustering-based color segmentation [25], [26] and morphological operations. The work further emphasizes the more accurate segmentation of overlapping cells. It consists of three crucial steps, including: 1) seed-point detection; 2) contour-evidence extraction; and 3) EF. We propose an efficient geometric EF approach motivated by the EFM of Prasad *et al.* [6].

A. Contribution

The major contributions of the proposed method are as follows.

- 1) Image quality improvement (edge enhancement, deblurring, and noise removal) using an Laplacian-of-Gaussian (LoG) [28], [29] based modified highboosting operation.
- 2) An efficient seed-point detection by employing bounded opening followed by fast radial symmetry (BO-FRS).
- 3) A novel geometric EF approach (where major and minor axes are accurately estimated based on the residue and residue-offset values).
4. Combining this geometric ellipse fitting with the algebraic EF, suggested by Zafari *et al.* [5], to achieve better performance by retaining the advantages of both the approaches.

This article is organized as follows. Section II highlights the proposed method for the segmentation of RBCs. Section III demonstrates the computational complexity of the proposed EFM. The performance evaluation is presented in Section IV. Finally, this article is concluded in Section V.

II. PROPOSED METHOD

The schematic of the proposed method for the precise segmentation of blood cells is illustrated in Fig. 1. It is composed of the following steps.

A. Preprocessing

The quality of the captured image suffers due to the presence of blur and unwanted noise, which may result a faulty diagnosis [1]. Thus, preprocessing is required to improve the

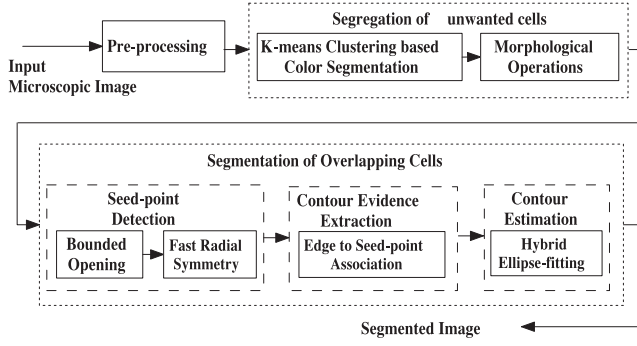


Fig. 1. Schematic of the proposed method.

image quality. Laplacian operator can be employed for edge enhancement and deblurring. However, it is sensitive to noise. Hence, to minimize the noise-effect, a 2-D Gaussian filter is employed prior to the Laplacian operator; making it a LoG filter [28]. It is also known as “Mexico hat.” Rotational symmetry is one of the salient features of LoG [29]. Here, we employ an LoG-based modified highboosting operation to produce a noise eliminated deblurred image. This operation is represented as follows:

$$F_I(x, y) = C_I(x, y) + kL_G(x, y) \quad (1)$$

where $C_I(x, y)$ depicts the input color image while $F_I(x, y)$ represents the preprocessed image. k is a weight factor ($k > 1$). The weight factor k should be properly selected to boost the image quality efficiently. Here, we take $k = 1.5$. $L_G(x, y)$ denotes LoG on $C_I(x, y)$, which is represented as

$$L_G(x, y) = \frac{-1}{\pi\sigma^4} \left(1 - \frac{x^2 + y^2}{2\sigma^2} \right) \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (2)$$

where σ symbolizes standard deviation, which should be properly selected to improve the image quality. Here, we employ an LoG filter of 7×7 with $\sigma = 1$.

B. Elimination of Unwanted Blood Cells

The image may have unwanted cells (WBCs and platelets in SCA detection; RBCs and platelets in AML and ALL detection), which may cause a false diagnosis [1]. Our proposed method emphasizes the elimination of unwanted cells using the k -means clustering-based color segmentation [25], [26] and morphological operations to improve segmentation accuracy and reliability. k -means clustering is an unsupervised machine learning technique [25]–[27]. It is employed to extract WBCs and platelets from an image, as highlighted in Pseudo Code 1. The detection of WBCs is a binary-class color segmentation problem, where each class is represented by three color channels. Hence, the number of clusters (n) in k -means clustering is selected as 6. Moreover, $n = 6$ is the optimal number of clusters at which WBCs are detected efficiently.

For the detection of ALL and AML, morphological operations are employed on (W) to eliminate the platelets like small particles and get the desired WBCs. However, in the SCA dataset to get the desired RBCs, we have first to convert (F_I) and (W) into HSV color space. Since the saturation component

Pseudo Code 1 Detection of WBCs and Platelets

Input: An RGB color image (F_I)

Output: An RGB color image having only WBCs and platelets **Begin**

1: Convert the input color image to L^*a^*b color space.

1: Initialize the number of clusters: $n = 6$.

3: k -means clustering is employed to classify color in “ a^*b^* ” space.

4: Level each pixel based on its cluster index.

5: Find out the resultant RGB color-image (W) .

End

(S) of RBCs and WBCs differ significantly, we subtract the S component of (W) from that of (F_I) to eliminate the detected WBCs and platelets. Then, apply morphological operations to completely remove WBCs, and platelets and get the S component of desired RBCs. Finally, the resultant RGB color image containing only RBCs are extracted successfully.

C. Segmentation of Overlapped Cells

This article presents a novel approach to efficiently segment cells, especially when a few cells overlap over each other as demonstrated in Pseudo Code 2. It consists of three crucial steps: 1) seed-point detection; 2) contour-evidence extraction [5]; and 3) contour estimation. In the following, we highlight these crucial steps for the segmentation of overlapped cells.

1) *Seed-Point Detection*: Seed-point detection is a crucial task in the segmentation of overlapping cells. It influences the performance of the final segmentation outputs. The main objective of seed-point detection is to identify each particle in an image by seed point. It also gives information about the total number of particles (cells) available in the image. Hence, a proper seed point detection boosts the performance of contour-evidence extraction as well as the contour estimation.

Zafari *et al.* [5] have suggested an effective seed-points detection approach based on BE followed by FRS [22]. It can efficiently detect seed points in silhouette images. However, the minor axis length gradually decreases due to BE. Hence, it is unable to properly detect the seed points of cells, which have a relatively large aspect ratio [1]. To mitigate this issue, we employ the bounded opening (BO) in place of BE, which can efficiently separate convex-shape cells. BO is simply an iterative opening process having a specific number of iteration. Here, the number of opening operations in BO is selected as four since it yields an optimal tread off between the seed-point detection performance and the number of iterations. FRS [22] is employed to detect seed regions having high rotational symmetry. Moreover, the proposed BO-FRS gives priority to both convexity and rotational symmetry of cells.

A gray-scale image G_I is represented as the union of convex cells C_j . Mathematically, G_I can be expressed as

$$G_I = \bigcup_{j=1}^n C_j. \quad (3)$$

The main motive of employing BO is to enhance the separability of each convex cell C_j . In the k th iteration of BO, every connected object C_j of $G_I(k-1)$ is processed to produce

resultant cell R_j . R_j is represented as

$$R_j = C_j^k \circ D(0, 1). \quad (4)$$

Here, $D(0, 1)$ represents the closed disc structuring object having unity radius and $\circ$ symbolizes morphological opening [21]. The resultant image in the k th iteration of BO is obtained as a union of all R_j

$$G_I^{k+1} = \bigcup_j R_j. \quad (5)$$

FRS is a computationally efficient feature extraction approach [22]. It concentrates on local radial symmetry (LRS) to extract seed points [5]. While LRS is employed with $N \times N$ neighborhood over the image having M pixels, it yields the complexity of order $O(MN)$ [22]. FRS is usually employed on the gradient of an image. Here, each edge pixel of an image delivers a vote to achieve the best LRS within a certain distance from this pixel [5]. We have to select an appropriate range of distance value d . Let d vary in $[R_{\min}, R_{\max}]$ for each pixel of gradient image I .

FRS emphasizes the estimation of magnitude projection image M_d and orientation projection image O_d . For this purpose, it first estimates positively affected A_{+ve} and negatively affected A_{-ve} pixels

$$A_{+ve}(x, y) = (x, y) + \text{round}\left(\left(\frac{I(x, y)}{\|I(x, y)\|}\right) \times d\right) \quad (6)$$

$$A_{-ve}(x, y) = (x, y) - \text{round}\left(\left(\frac{I(x, y)}{\|I(x, y)\|}\right) \times d\right) \quad (7)$$

$$O_d(A_{+ve}(x, y)) \rightarrow O_d(A_{+ve}(x, y)) + 1 \quad (8)$$

$$O_d(A_{-ve}(x, y)) \rightarrow O_d(A_{-ve}(x, y)) - 1 \quad (9)$$

$$M_d(A_{+ve}(x, y)) = M_d(A_{+ve}(x, y)) + \|I(x, y)\| \quad (10)$$

$$M_d(A_{-ve}(x, y)) = M_d(A_{-ve}(x, y)) - \|I(x, y)\|. \quad (11)$$

The contribution of radial symmetry S_d is estimated as

$$S_d = \frac{M_d(x, y)}{k_d} \left(\frac{|\tilde{O}_d(x, y)|}{k_d} \right)^\lambda * H_d \quad (12)$$

where $*$ symbolizes convolution operation and H_d denotes 2-D Gaussian. Here, λ is the radial strictness and k_d is a scaling-factor responsible for the normalization of M_d and O_d . $\tilde{O}_d$ is represented as

$$\tilde{O}_d(x, y) = \begin{cases} O_d(x, y), & \text{if } O_d(x, y) < k_d \\ k_d, & \text{otherwise.} \end{cases} \quad (13)$$

The full FRS-transform is computed as the mean of the symmetry contributions over all the radii $d \in [R_{\min}, R_{\max}]$

$$S = \frac{1}{|N|} \sum_{d \in [R_{\min}, R_{\max}]} S_d. \quad (14)$$

Finally, the seed points are predicted based on average location, centroid, of the identified symmetry regions, in S .

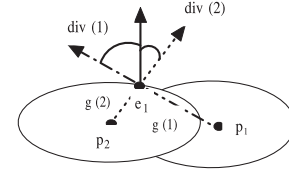


Fig. 2. Example of edge-to-seed-point association.

2) *Contour-Evidence Extraction*: In this section, we emphasize the detection of contour evidence based on the edge-to-seed point association technique [30]. This technique can successfully assign each edge pixel to the corresponding seed point based on the Euclidean distance and the divergence factor. For this purpose, we first extract the edge points based on the Sobel edge detection technique [20], which is focused on high spatial frequency to identify the appropriate edge points. For example, let the set of detected seed points be $P = \{p_1, p_2, \dots, p_n\}$. Each edge point e_j in $E = \{e_1, e_2, \dots, e_r\}$ is associated to the corresponding seed points depending on a relevance factor $\text{rel}(e_j, p_k)$ given as

$$\text{rel}(e_j, p_k) = \frac{1 - \omega}{1 + \text{dist}(e_j, p_k)} + \omega \frac{\text{div}(e_j, p_k) + 1}{2} \quad (15)$$

where ω is a constant taking on values $[0, 1]$. Here, $\text{dist}(e_j, p_k)$ is the normalized Euclidean distance between e_j and p_k , and $\text{div}(e_j, p_k)$ denotes the divergence function between e_j and p_k . Both the terms lie within $(0, 1]$. The edge pixel e_j is associated with a seed point p_k only when relevance value between them is maximum.

It is considered that each pixel on the line joining the edge point-to-seed point $l(e_j, p_k)$ should lie in the image foreground I_f

$$g(x) = \begin{cases} |e_j - p_k|, & \text{if } l(e_j, p_k) \subset I_f \\ \infty, & \text{otherwise.} \end{cases} \quad (16)$$

The deviation between the direction of $l(e_j, p_k)$ and the gradient direction at e_j is represented by $\text{div}(e_j, p_k)$. It is expressed as

$$\text{div}(e_j, p_k) = \frac{l(e_j, p_k)g(e_j)}{\|l(e_j, p_k)\| \|g(e_j)\|}. \quad (17)$$

The search space is adaptively determined through a circular region around every edge point. Hence, less number of seed points are processed, and erroneous edge points are eliminated successfully. Moreover, it makes the edge-to-seed-point association more robust and computationally more efficient.

Fig. 2 depicts the edge-to-seed-point association. Here, the dashed arrow illustrates the line from seed point to edge point. The solid arrow indicates the gradient direction at e_j .

3) *Contour Estimation*: The prime objective of this section is to segment overlapping cells using the contour estimation properly. Here, we propose a novel geometrical EF approach inspired by the EF technique of Prasad *et al.* [6]. It is a noniterative, LS-based EF technique. Moreover, the major and minor axes are updated more precisely based on the residue and residue-offset factors. The geometric equation of an ellipse

is represented as

$$\frac{((x - x_0) \cos \theta_0 - (y - y_0) \sin \theta_0)^2}{m^2} + \frac{((x - x_0) \sin \theta_0 + (y - y_0) \cos \theta_0)^2}{n^2} = 1 \quad (18)$$

where

- m : semimajor axis length;
- n : semiminor axis length;
- θ_0 : angle of orientation (angle between x -axis and major axis);
- $C(x_0, y_0)$: center equation (18) represents an ellipse, provided the following conditions are satisfied:

$$\begin{aligned} C1 : m, n &\in R^+ \\ C2 : m &\geq n \\ C3 : \theta_0 &\in [0, \pi) \\ C4 : x_0, y_0 &\in R. \end{aligned} \quad (19)$$

Equation (18) can be rewritten as

$$\alpha(x - x_0)^2 + \beta(y - y_0)^2 + \gamma(x - x_0)(y - y_0) = (mn)^2 \quad (20)$$

where

$$\begin{aligned} \alpha &= 0.5 \left((m^2 + n^2) - (m^2 - n^2) \cos(2\theta_0) \right) \\ \beta &= 0.5 \left((m^2 + n^2) + (m^2 - n^2) \cos(2\theta_0) \right) \\ \gamma &= (m^2 - n^2) \sin(2\theta_0). \end{aligned} \quad (21)$$

The mathematical model of EF is demonstrated as follows.

Suppose $\bar{E}$ is a five-dimensional vector, which consists of ellipse parameters: m, n, θ_0, x_0, y_0 , the vector utilizing the sequence of pixels $P_k'(x_k', y_k')$: $k = 1 : N$. The vector $\bar{Y}$ contains y -coordinates of the pixels

$$\bar{E} = [m, n, \theta_0, x_0, y_0]^T \quad (22)$$

$$\bar{Y} = [-y_1^2 \quad -y_2^2 \quad \dots \quad -y_N^2]^T, \quad \bar{Y} \in Z^N \quad (23)$$

$$\bar{H} = [h_1 \quad h_2 \quad h_3 \quad h_4 \quad h_5]^T, \quad \bar{H} \in R^5 \quad (24)$$

$$G : \bar{E} \rightarrow \bar{H}; X : \bar{H} \rightarrow \bar{Y}. \quad (25)$$

$\bar{H}$ is a five-dimensional vector, which contains new real-valued variables h_1 to h_5 . It is employed to split the nonlinear mapping $F : \bar{E} \rightarrow \bar{Y}$ into two maps G and X as given by (25). The variables h_1 to h_5 are estimated in such a manner that it yields linear mapping between $\bar{H}$ and $\bar{Y}$, represented by

$$\bar{Y} = X\bar{H}. \quad (26)$$

$G : \bar{E} \rightarrow \bar{H}$ is a nonlinear map whereas the mapping between $\bar{H}$ and $\bar{Y}$ is linear; and thus maintaining the mapping of $\bar{E}$ to $\bar{Y}$ is nonlinear. However, the parameters of $\bar{H}$ and $\bar{E}$ make G as a one-to-one mapping. Thus, m, n, θ_0, x_0, y_0 can be evaluated distinctively from h_1 to h_5 . For evaluating the parameters h_1 to h_5 and constructing the map X , we characterize residual distance as

$$\sum_{k=1}^N r_k^2 = \|\bar{Y} - X\bar{H}\|^2 \quad (27)$$

where $\|\cdot\|$ symbolizes the Euclidean norm of a vector. This residual distance is minimized to obtain the unique optimal solution of $\bar{H}$. $\bar{H}$ can be estimated as

$$\bar{H} = (X^T X)^{-1} X^T \bar{Y}. \quad (28)$$

The ellipse formulation, given in (20), may be rewritten as

$$\begin{aligned} -y^2 &= \frac{\alpha}{\beta} (x^2) + \frac{\gamma}{2\beta} (2xy) - 2x \left(\frac{\alpha x_0 + \frac{\gamma}{2} y_0}{\beta} \right) \\ &\quad - 2y \left(\frac{\beta y_0 + \frac{\gamma}{2} x_0}{\beta} \right) \\ &\quad - \left(\frac{m^2 n^2 - (\alpha x_0^2 + \beta y_0^2 + \gamma x_0 y_0)}{\beta} \right). \end{aligned} \quad (29)$$

The parameters h_1 to h_5 , and the map X can be evaluated by solving (29) as follows:

$$h_1 = \alpha / \beta \quad (30)$$

$$h_2 = \gamma / 2\beta \quad (31)$$

$$h_3 = \frac{2\alpha x_0 + \gamma y_0}{2\beta} = h_1 x_0 + h_2 y_0 \quad (32)$$

$$h_4 = \frac{2\beta y_0 + \gamma x_0}{2\beta} = y_0 + h_2 x_0 \quad (33)$$

$$h_5 = \frac{m^2 n^2 - (\alpha x_0^2 + \beta y_0^2 + \gamma x_0 y_0)}{\beta} \quad (34)$$

$$= \frac{m^2 n^2}{\beta} - (h_1 x_0^2 + y_0^2 + 2h_2 x_0 y_0)$$

$$X = \begin{bmatrix} x^2 & 2xy & -2x & -2y & -1 \end{bmatrix}. \quad (35)$$

The normalized residue is estimated as

$$R_n = \frac{\|X\bar{H} - \bar{Y}\|}{\|\bar{Y}\|} \quad (36)$$

where $\|\cdot\|$ depicts the Euclidean norm of a vector. Perfect EF can be achieved at a minimal normalized residue. The normalized residue varies between 0 and 1.

$$m = \sqrt{\frac{2 \left((h_2 + h_5)(h_2^2 - h_1^2) + (h_2 h_3 - h_1 h_4)^2 + h_1(h_2 h_4 - h_3^2) \right)}{(h_2^2 - h_1^2) \left(1 + h_1 + 1.5R_n - \sqrt{(1 - h_1 - R_n')^2 + 4(h_2 - 0.5R_n')^2} \right)}} \quad (37)$$

$$n = \sqrt{\frac{2 \left((h_2 + h_5)(h_2^2 - h_1^2) + (h_2 h_3 - h_1 h_4)^2 + h_1(h_2 h_4 - h_3^2) \right)}{(h_2^2 - h_1^2) \left(1 + h_1 + 1.5R_n + \sqrt{(1 - h_1 - R_n')^2 + 4(h_2 - 0.5R_n')^2} \right)}} \quad (38)$$

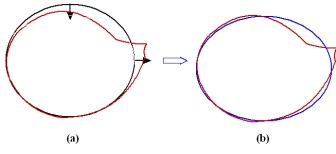


Fig. 3. Red line represents the contour of an object. (a) Black line represents the EF without considering the terms R_n and R'_n in (37) and (38). (b) Blue line represents the EF by considering R_n and R'_n .

The parameters of the ellipse: m, n, θ_0, x_0, y_0 can be evaluated from h_1 to h_5 as shown in (37)–(41), respectively

$$\theta_0 = -0.5 \tan^{-1} \left(\frac{2h_2}{1-h_1} \right) \quad (39)$$

$$x_0 = \frac{h_3 - h_2 h_4}{h_1 - h_2^2} \quad (40)$$

$$y_0 = \frac{h_1 h_4 - h_2 h_3}{h_1 - h_2^2} \quad (41)$$

where the term R'_n , the *residue offset* in (37) and (38), shown the bottom of the previous page, is estimated as

$$R'_n = \text{mean}(\text{sign}(X\bar{H} - \bar{Y}))R_n. \quad (42)$$

The terms R_n and R'_n are responsible for fine-tuning and precise fine-tuning (the EF), respectively. The term R_n modifies the minor and major axes to provide better EF. Furthermore, R'_n optimizes the major and minor axes depending upon the mean $(\text{sign}(X\bar{H} - \bar{Y}))$, as shown in Fig. 3. In this example, $\text{mean}(\text{sign}(X\bar{H} - \bar{Y}))$ is negative, which makes R'_n negative. Hence, it optimizes the EF by increasing major-axis length and decreasing the minor-axis length based on R'_n . Similarly, positive $\text{mean}(\text{sign}(X\bar{H} - \bar{Y}))$ enhances the minor-axis (n) and reduces the major-axis (m); thus yielding a precise fine-tuning of EF.

Usually, the above-proposed EFM demonstrates excellent performance. However, sometimes it may display relatively poor performance particularly, when $R_n > 0.45$, $m > \rho$, $n > \rho$. Specifically, in such situations, we employ the LS-based noniterative algebraic EFM proposed by Zafari *et al.* [5] to improve the performance and make the EF more accurate. We take $\rho = 30$ for SCA and ALL datasets, whereas $\rho = 60$ for AML dataset. Hence, the proposed method combines the benefits of both geometric and algebraic EF techniques.

III. COMPUTATIONAL COMPLEXITY OF PROPOSED ELLIPSE-FITTING APPROACH

The computational complexity of the proposed hybrid EFM is estimated, as mentioned in the following.

- 1) The computational complexity for finding $X^T X$ is $O(12N)$ [6].
- 2) The computational complexity for estimating $(X^T X)^{-1}$ from $X^T X$ is $O(5^3)$ [6].
- 3) The computational complexity for evaluating $\bar{H}$ is $O(30N)$ [6].
- 4) The computational complexity for computing R_n is $O(14N)$.
- 5) The computational complexity for determining R'_n is $O(N)$.

Pseudo Code 2 Segmentation of Overlapping Cells

Input: The resultant color image after removal of unwanted cells

Output: Final segmented color image

Begin

1: Initialization of parameters

Set number of opening operations in BO as 4.

Set radial range:

$[R_{MIN}, R_{MAX}] = [12, 21]$ for SCA and ALL datasets
 $= [14, 31]$ for AML dataset.

2: Detect the seed-points by employing BO-FRS.

3: Extract the contour evidence using edge-to-seed-point association.

4: Construct the matrix X and $\bar{Y}$ using Eq. (35) and (23), respectively.

5: Compute $\bar{H}$ utilizing Eq. (28).

6: Evaluate the normalized residue (R_n) using Eq. (36).

7: Estimate m, n, θ_0, x_0, y_0 employing Eq. (37)–(41), respectively.

8: To make the ellipse-fitting more efficient, employ the suggested algebraic ellipse-fitting particularly when $R_n > 0.5$, $m > \rho$, $n > \rho$.

End

6) The computational complexity for determining $\bar{E}$ is $O(5)$.

7) The computational complexity of the employed algebraic EF approach is $\sim O(16M)$.

The computational complicity of the proposed EF approach is estimated mostly by the computational complexity of $X^T X$, $\bar{H}$, $\bar{R}_n$, $\bar{R}'_n$, and algebraic EF approach. Thus, the overall computational complexity is $O(57N) + O(16M)$. Since only in some specific situations ($R_n > 0.45$, $m > \rho$, $n > \rho$), we employ the LS-based noniterative algebraic EFM proposed by Zafari *et al.* [5], M is very small than N . Hence, the proposed hybrid EF approach is computationally efficient since its computational complexity is $\sim O(N)$.

IV. PERFORMANCE EVALUATION

In this section, we present the performance of the proposed method as well as various existing methods for the segmentation of cells. Next, we discuss the datasets and the performance measures used. Finally, the performance analysis of the proposed method is presented.

A. Dataset

The proposed method is validated using the following publicly available datasets: SCA dataset [ErythrocytesIDB [31]], ALL dataset (ALLIDB1 [32]), and AML dataset [33].

B. Performance Measures

We emphasize various performance measures: precision [1], recall [1], F1 score [1], Jaccard coefficient (J_c) [5], and Dice similarity coefficient (DSC) [35] for quantitative performance analysis.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (43)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (44)$$

$$F1 \text{ Score} = \frac{2 \times (\text{Recall} \times \text{Precision})}{(\text{Recall} + \text{Precision})} \quad (45)$$

$$J_c = \frac{|I_s \cap I_g|}{|I_s \cup I_g|}. \quad (46)$$

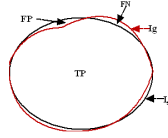


Fig. 4. Confusion matrix venn diagram.

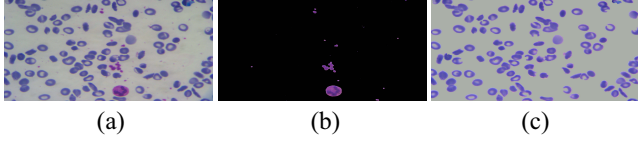


Fig. 5. Segmentation of unwanted cell in the sickle-cell dataset. (a) Original image. (b) Extracted unwanted cells. (c) Resultant image having only RBCs.

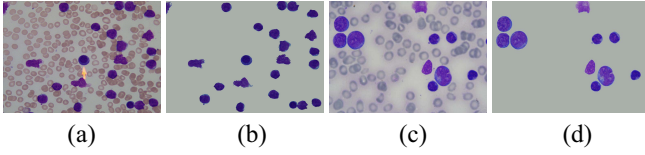


Fig. 6. Segmentation of unwanted cell in ALL and AML datasets: (a) and (c) original images; (b) and (d) resultant images containing only WBCs.

$$DSC = \frac{2|I_s \cap I_g|}{|I_s| + |I_g|} \quad (47)$$

where I_s represents segmented image and I_g symbolizes ground-truth image. True positive (TP) characterizes the number of properly segmented pixels. False negative illustrates the number of undetected pixels. False positive (FP) is the number of falsely identified pixels, which truly are not so, as shown in Fig. 4. The ideal value of DSC, (J_c), and (A_s) is 1.

C. Results

This section presents the performance of the proposed segmentation method for accurate detection of hematological disorders. We emphasize the performance evaluation of the elimination of unwanted cells, seed-point detection, and segmentation of overlapping cells. Moreover, the performance of the proposed EF approach is compared with other existing methods, including EFM of Zafari *et al.* [5], Prasad *et al.* [6], and Meng *et al.* [7]. Moreover, we apply the same pipeline structure as presented in the proposed method to make the comparison more fair and effective.

1) *Elimination of WBCs and Platelets*: In this section, we illustrate the qualitative performance of the elimination of unwanted cells. Fig. 5(a) depicts the input image, which contains RBCs along with WBCs and platelets. These unwanted cells are extracted successfully by employing the k-means clustering-based color segmentation technique, as shown in Fig. 5(b). Fig. 5(c) illustrates the desired resultant image, which contains only RBCs. From the figure, we visualize that WBCs and platelets are eliminated perfectly, which will make the segmentation more accurate. Moreover, from Fig. 6(b) and (d), we observe that WBCs are extracted successfully from ALL and AML dataset images, respectively.

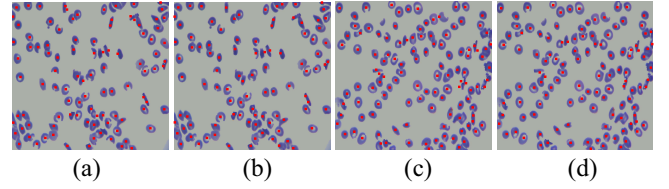


Fig. 7. Seed point detection: (a) and (c) BE-FRS; (b) and (d) BOFRS.

TABLE II
PERFORMANCE COMPARISON OF SEED-POINT DETECTION

Dataset	Method	Precision (%)	Recall (%)	F1 Score (%)
Sickle cell [31]	BEFRS	95.80	95.00	95.40
	BOFRS	96.64	95.83	96.23
ALL [32]	BEFRS	98.63	98.18	98.40
	BOFRS	99.09	98.64	98.86
AML [33]	BEFRS	99.00	98.5	98.75
	BOFRS	99.33	99.00	99.16

2) *Seed-Point Detection*: The prime objective of seed-point detection is to identify the location of the cells properly. Fig. 7 demonstrates the qualitative performance of seed-point detection. BO-FRS can detect seed points more precisely than BE-fast radial symmetry (BE-FRS). From these figures, we observe that BE-FRS illustrates comparatively poor-performance due to false detection of seed points, particularly the cells having a large aspect ratio. Table II illustrates that BOFRS demonstrates superior performance in terms of precision, recall, and F1 score as compared to BEFRS.

3) *Segmentation of Overlapping Cells*: In this section, we present both the qualitative and quantitative performance of the segmentation of cells using EF approaches. The proposed hybrid EF technique is compared with various existing methods, including the EF techniques suggested by Zafari *et al.* [5], Prasad *et al.* [6], and Meng *et al.* [7]. Furthermore, the proposed method is compared with the deep-learning-based semantic segmentation approaches like U-Net [37] and DeepLab [38] to demonstrate the proposed method's superiority. For this purpose, we split the dataset as 60% for training, 20% for validation, and rest for testing. For fair comparative analysis, the testing performances in each of the seven experiments are demonstrated, where, in each experiment, the corresponding image is only used for testing.

Fig. 8 demonstrates the qualitative performance of the segmentation of cells in the SCA dataset [31]. From these figures, we visualize that the proposed EF technique yields the best performance among all with proper detection of cells. However, Zafari *et al.* [5] depicted the second-best performance.

However, on a rare occasion, the proposed hybrid EF approach demonstrates the poor segmentation performance, particularly for very few cells, as shown in Fig. 8. From the figure, we observe that it imprecisely segment the two cells (one cell from the overlapping cells and the other cells near to it, which lies in the middle of the uppermost row in Experiment 1). It occurs due to the low breakdown point limitation since both the proposed geometric and the employed algebraic approaches are LS-based and noniterative in nature. Moreover, in such a situation (for these two cells), the

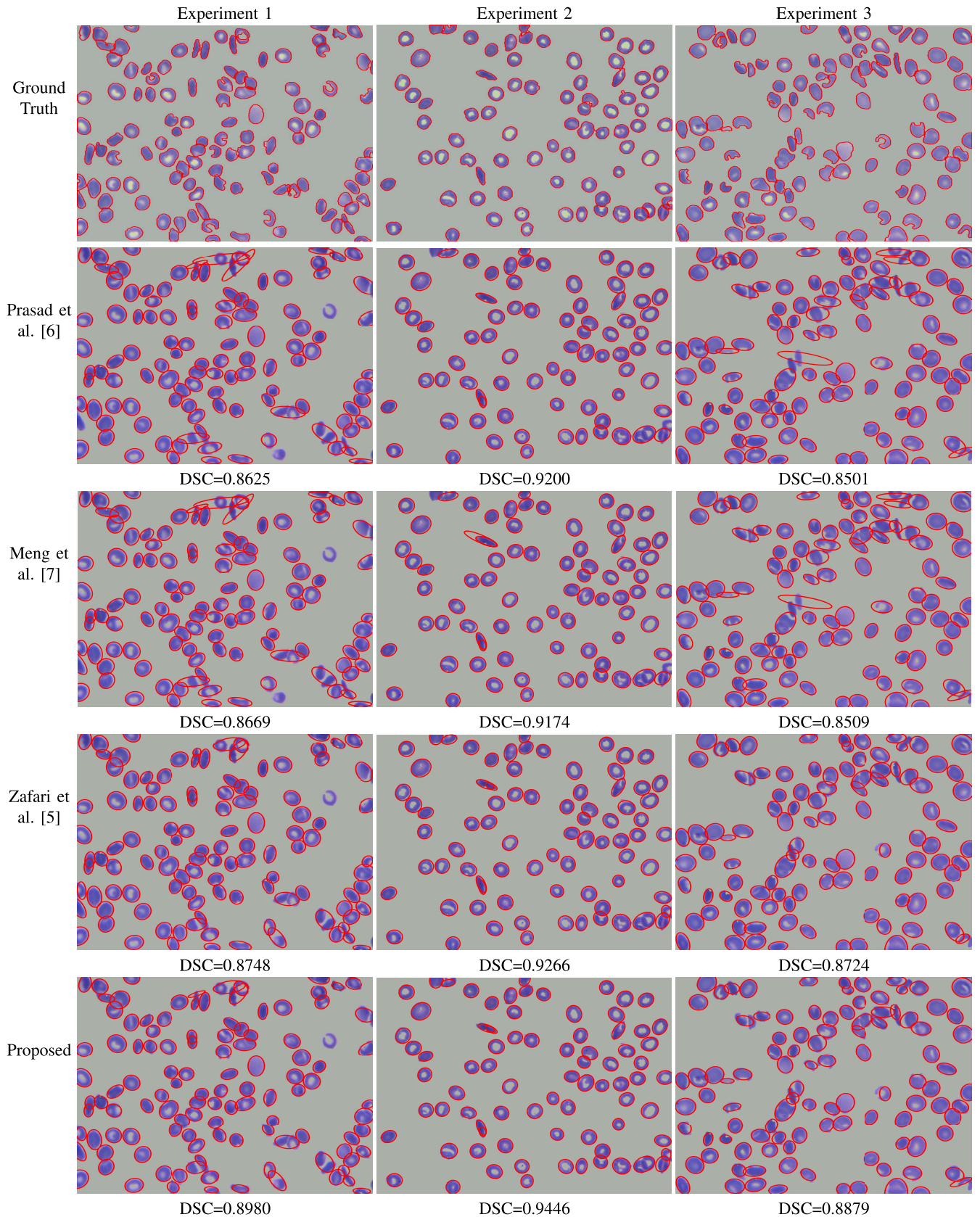


Fig. 8. Segmentation of cells in the SCA dataset [31].

proposed method and the EFM suggested by Zafari *et al.* [5] outperform others with superior performance. Furthermore, the hybridization of noniterative LS-based geometric and algebraic

approaches makes the proposed method computationally efficient. In Experiment 2, all these EFMs, including the proposed method, yield relatively poor performance, specifically when

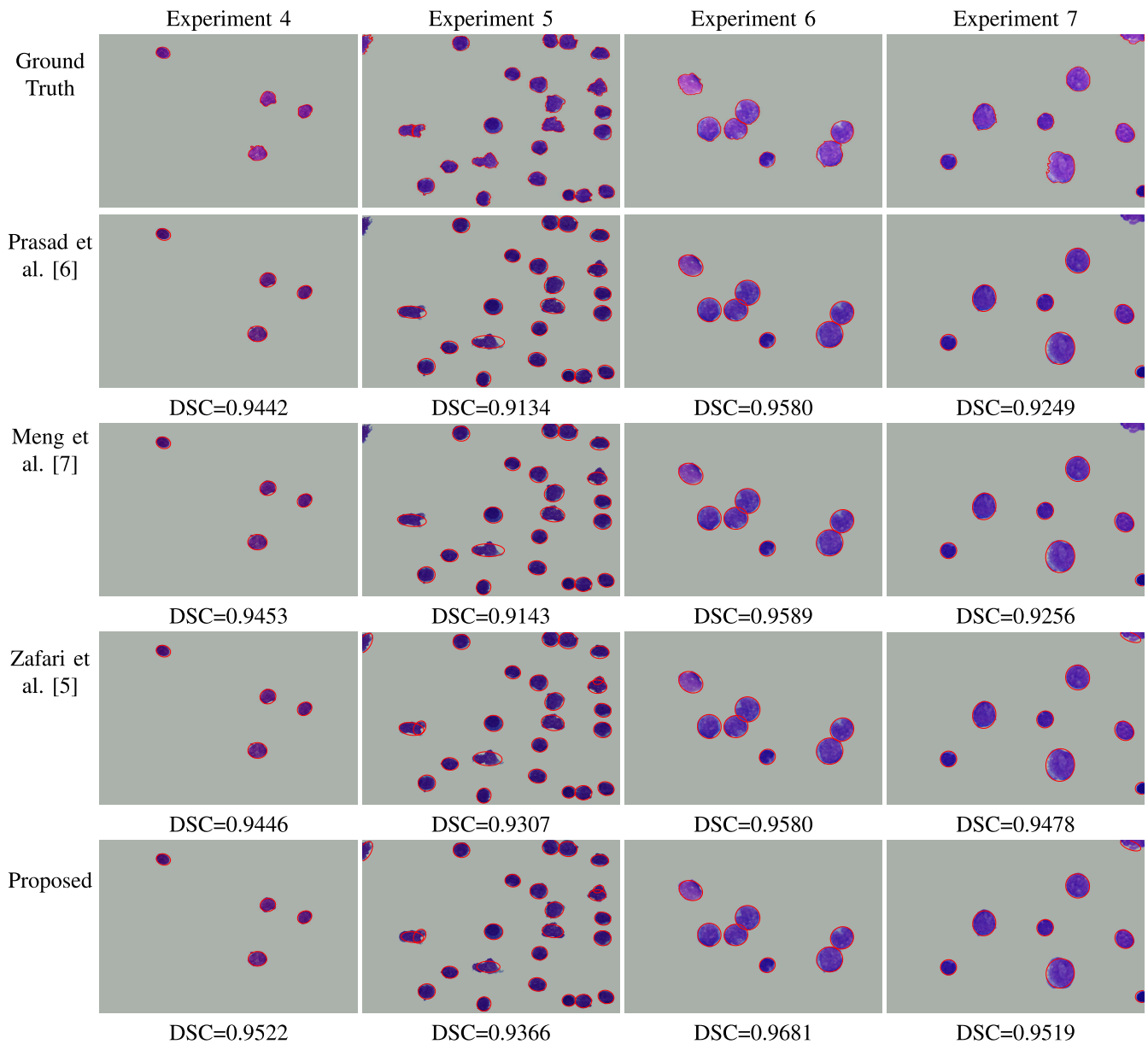


Fig. 9. Segmentation of cells in Leukemia datasets [32], [33].

two elongated cells lie in the middle due to false detections of one extra seed point.

In Fig. 9, Experiments 4 and 5 represent the qualitative performance of segmentation of cells in ALL dataset [32], whereas Experiments 6 and 7 demonstrate the performance in the AML dataset [33]. From the figure, we observe that the proposed EF approach yields outstanding performance in all these experiments since it retains the benefits of both the geometric and algebraic approaches.

In Experiment 4, the EF approach presented by Meng *et al.* [7] yields the second-best performance, whereas the EF technique suggested by Zafari *et al.* [5] ranked second in Experiment 5. Experiments 6 and 7 illustrate all these EF technique yields excellent performance. Moreover, the proposed method outperforms with proper segmentation of cells.

The quantitative performance of segmentation of cells in SCA [31], ALL [32], and AML [33] datasets are shown in Tables III–V, respectively. From these tables, we notice that the proposed method yields superior performance with the best DSC, Jaccard score, precision, and F1 score. In all the experiments except Experiments 4 and 6, the EF technique suggested by Zafari *et al.* [5] ranked second. On the other hand, in Experiments 4 and 6, the EF approach presented by Meng *et al.* [7] demonstrated the second-best performance. Moreover, in all these seven experiments, the proposed hybrid EF technique outperforms with the best DSC, Jaccard score, precision, and F1 score since it retains the benefits of both the geometric and algebraic methods. Furthermore, the efficient estimation of minor and major axes based on residue and residue offset helps it to achieve the best performance.

TABLE III
PERFORMANCE COMPARISON USING THE SCA DATASET [31]

Ex.	Method	Precision (%)	F1 Score (%)	J_c	DSC
1	U-Net [37]	70.32	81.01	0.6957	0.8101
	DeepLab [38]	77.54	86.61	0.7602	0.8661
	Prasad et. al EFM [6]	79.36	86.25	0.7582	0.8625
	Meng et al. EFM [7]	79.69	86.69	0.7651	0.8669
	Zafari et al. EFM [5]	80.47	87.48	0.7775	0.8748
	Proposed EFM	85.97	89.81	0.8150	0.8597
2	U-Net [37]	73.06	83.24	0.7185	0.8324
	DeepLab [38]	79.20	88.10	0.7845	0.8810
	Prasad et. al EFM [6]	86.63	92.00	0.8519	0.9200
	Meng et al. EFM [7]	86.19	91.74	0.8474	0.9174
	Zafari et al. EFM [5]	86.71	92.66	0.8632	0.9266
	Proposed EFM	91.67	94.46	0.8950	0.9446
3	U-Net [37]	70.51	80.89	0.6960	0.8089
	DeepLab [38]	76.73	86.26	0.7628	0.8626
	Prasad et. al EFM [6]	75.90	85.01	0.7393	0.8501
	Meng et al. EFM [7]	76.07	85.09	0.7405	0.8509
	Zafari et al. EFM [5]	78.86	87.24	0.7737	0.8724
	Proposed EFM	83.82	88.79	0.7984	0.8879

TABLE IV
PERFORMANCE COMPARISON USING ALL DATASET [32]

Ex.	Method	Precision (%)	F1 Score (%)	J_c	DSC
4	U-Net [37]	81.32	88.30	0.8038	0.8830
	DeepLab [38]	88.93	93.67	0.8817	0.9367
	Prasad et. al EFM [6]	90.55	94.42	0.8943	0.9442
	Meng et al. EFM [7]	90.87	94.53	0.8963	0.9453
	Zafari et al. EFM [5]	90.54	94.46	0.8950	0.9446
	Proposed EFM	95.67	95.22	0.9088	0.9522
5	U-Net [37]	80.07	87.78	0.7925	0.8778
	DeepLab [38]	87.26	91.88	0.8649	0.9188
	Prasad et. al EFM [6]	88.40	91.34	0.8406	0.9134
	Meng et al. EFM [7]	88.61	91.43	0.8421	0.9143
	Zafari et al. EFM [5]	88.44	93.07	0.8704	0.9307
	Proposed	94.02	93.66	0.8808	0.9366

TABLE V
PERFORMANCE COMPARISON USING THE AML DATASET [33]

Ex.	Method	Precision (%)	F1 Score (%)	J_c	DSC
6	U-Net [37]	69.72	77.56	0.6845	0.7756
	DeepLab [38]	74.31	81.47	0.7314	0.8147
	Prasad et. al EFM [6]	92.40	95.80	0.9194	0.9580
	Meng et al. EFM [7]	92.61	95.89	0.9210	0.9589
	Zafari et al. EFM [5]	92.39	95.80	0.9194	0.9580
	Proposed EFM	95.84	96.81	0.9382	0.9681
7	U-Net [37]	71.09	78.75	0.7012	0.7875
	DeepLab [38]	75.13	82.06	0.7397	0.8206
	Prasad et. al EFM [6]	93.09	92.49	0.8603	0.9249
	Meng et al. EFM [7]	93.30	92.56	0.8615	0.9256
	Zafari et al. EFM [5]	92.67	94.78	0.9008	0.9478
	Proposed EFM	96.15	95.19	0.9082	0.9519

We compare the proposed method with efficient deep-learning-based semantic segmentation approaches like U-Net [37] and DeepLab [38] along with the EF approaches to make the performance analysis more attractive. Since the standard datasets: SCA (ErythrocytesIDB [31]), ALL (ALLIDB1 [32]), and AML [33] are small datasets, deep-learning methods show relatively poor performance. From the experimental results, we observe that the DeepLab network outperforms U-Net [37] due to its salient contributions: 1) atrous convolution and 2) atrous-spatial pyramid pooling. Atrous convolution is responsible for updating the resolution at which feature responses are estimated. On the other hand, atrous-spatial pyramid pooling is employed to depict more robust multiscale object segmentation. In

TABLE VI
COMPUTATIONAL TIME FOR EF(S)

Dataset	Prasad et. al [6]	Meng et al. [7]	Zafari et al. [5]	Proposed
SCA [31] (Ex. 1-3)	3.59 s	5.84 s	3.64 s	3.68 s
ALL [32] (Ex. 4-5)	1.02 s	2.93 s	1.06 s	1.11 s
AML [33] (Ex. 6-7)	1.09 s	3.12 s	1.10 s	1.14 s

the DeepLab [38] network, the fully connected conditional random field responses and the final deep-CNN are integrated to yield more accurate object localization. However, it still depicts relatively poor performance, particularly due to false detection of background pixels as object pixels. Hence, it results in poor precision, F1 score, Jaccard coefficient, and DSC performances due to more false positives. On the other hand, our proposed method outperforms others by retaining the benefits of both geometric and algebraic approaches. Moreover, the introduced residue-offset factor improves segmentation performances with more precise EF. Furthermore, the proposed method is computationally more efficient than the deep-learning methods, and it does not require the training images to learn, unlike deep-learning methods. In AML datasets, U-Net [37] and DeepLab [38] based segmentation approaches display relatively weak performance since the AML dataset is very small, consisting only of 27 images.

4) *Computation Time*: The simulation experiment is carried out on a MATLAB R2019b platform operating on a hardware computing device comprising an Intel Core i5 processor clocked at 3.40 GHz and a 12-GB RAM. Table VI highlights the computation time required for the EF. From the table, we notice that the algorithms of Zafari *et al.* [5], Prasad *et al.* [6], and the proposed hybrid EF approach yields similar performance regarding computational time. However, the proposed method is faster than the EFM of Meng *et al.* [7]. Moreover, the proposed hybrid-ellipse fitting is preferred over others since it yields outstanding performance with comparable computational time.

V. CONCLUSION

Unlike the general segmentation of blood-cells algorithms, which are mostly based on the watershed segmentation, the segmentation algorithm discussed in this article is based on the BO-FRS-based seed-point detection and the hybrid EF-based contour estimation. The proposed methods have the ability to extract the seed points more accurately and to segment the overlapping cells more precisely, even if from the low-contrast inhomogeneous visual features. Therefore, the method is suitable for solving complex blood cell segmentation problems. The LS-based geometric ellipse fitting approach adopted in this article helps in more localization, therefore better accuracy. Usually, in algebraic EFMs, the fitting parameters are biased, and the fitting errors are unwillingly weighted, which leads to inaccurate boundaries, leading to problems in the detection of hematological disorders. On the other hand, our proposed method performs better than the existing methods because of the combined benefits of geometric and algebraic EFMs. Moreover, it detects the cells more accurately due to proper EF. It is a computationally efficient technique since

it hybridizes noniterative-geometric and algebraic methods. Furthermore, the noise problem is solved by employing an LoG-based modified highboosting operation. Hence, it is less sensitive to noise. Unlike other methods, our scheme never faces oversegmentation problem. Finally, we believe that it can also be employed for the segmentation of cells in other medical applications like- MRI, CT, ultrasound, and X-ray images, cybernetics applications, the segmentation of overlapping objects, etc. Last but not the least, the proposed algorithm does not need any training and hence does not need any training data; making it a very suitable approach than a deep-learning-based technique when there is very less or no data available for training.

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